RELATIVE PERFECTNESS IS LOCAL FOR THE ETALE TOPOLOGY ON HENSELIAN PAIRS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 13181-13261, immediately following the statement at 13170-13179.

Editorial note. Same-name pair note: this is the more general (algebraic spaces) side; the schemes side is not counted beside it.

Editorial note. Same-chapter pointers (characterisation, thickenings) are recorded in assembly.external_references.

Editorial note. Steps the author leaves to the reader are flagged here for the reviewer: source line(s) 13234 carry an "omitted"/"follows easily"/"immediate from the definitions" phrase. Those sentences are retained verbatim; the displayed argument around them stays inside the counted span.

Allowed standard prerequisites (editorial).

- Characterisation of relatively perfect morphisms. chapter lemma lemma-characterize-relatively-perfect, cited in the proof. Supplies the criterion that is checked etale-locally.
- Thickenings and pseudo-coherence. chapter lemmas lemma-thickening-pseudo-coherent and lemma-thickening-relatively-perfect, cited in the proof. Control the behaviour of the property along the thickenings appearing in the henselian setup.
- Henselian pairs and etale localization. the notion of a henselian pair used in the statement together with the etale-local reduction in the proof. Reduces the verification to the affine henselian case where the criterion applies.

Source reference. Original reference 0B7W: focused-seeds/geometry/S9-spaces-henselian-relatively-perfect-spaces/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03K2: focused-seeds/geometry/S9-spaces-henselian-relatively-perfect-spaces/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0DJH: focused-seeds/geometry/S9-spaces-henselian-relatively-perfect-spaces/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0DJF: focused-seeds/geometry/S9-spaces-henselian-relatively-perfect-spaces/ref-tags.txt, source lines 8–8. Navigation pointer only: the

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Source reference. Original reference 06Z0: focused-seeds/geometry/S9-spaces-henselian-relatively-perfect-spaces/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 08E7: focused-seeds/geometry/S9-spaces-henselian-relatively-perfect-spaces/ref-tags.txt, source lines 10–10. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0DJJ: focused-seeds/geometry/S9-spaces-henselian-relatively-perfect-spaces/ref-tags.txt, source lines 11–11. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 071Q: focused-seeds/geometry/S9-spaces-henselian-relatively-perfect-spaces/ref-tags.txt, source lines 12–12. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 08HE: focused-seeds/geometry/S9-spaces-henselian-relatively-perfect-spaces/ref-tags.txt, source lines 13–13. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 08JL: focused-seeds/geometry/S9-spaces-henselian-relatively-perfect-spaces/ref-tags.txt, source lines 14–14. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 04K5: focused-seeds/geometry/S9-spaces-henselian-relatively-perfect-spaces/ref-tags.txt, source lines 15–15. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let (R, I) be a pair consisting of a ring and an ideal I contained in the Jacobson radical. Set $S = \mathrm{Spec}(R)$ and $S_0 = \mathrm{Spec}(R/I)$. Let X be an algebraic space over R whose structure morphism $f : X \to S$ is proper, flat, and of finite presentation. Denote $X_0 = S_0 \times_S X$. Let $E \in D(\mathcal{O}_X)$ be pseudo-coherent. If the derived restriction E_0 of E to X_0 is S_0-perfect, then E is S-perfect.*

Proof. Choose a surjective étale morphism $U \to X$ with U affine. Choose a closed immersion $U \to \mathbf{A}^d_S$. Set $U_0 = S_0 \times_S U$. The complex $E_0|_{U_0}$ has tor amplitude in $[a, b]$ for some $a, b \in \mathbf{Z}$. Let $\overline{x}$ be a geometric point of X. We will show that the tor amplitude of $E_{\overline{x}}$ over R is in $[a - d, b]$. This will finish the proof as the tor amplitude can be read off from the stalks by Cohomology on Sites, Lemma 0DJJ and Properties of Spaces, Theorem 04K5.

Let $x \in |X|$ be the point determined by $\overline{x}$. Recall that $|X| \to |S|$ is closed (by definition of proper morphisms). Since I is contained in the Jacobson radical, any nonempty closed subset of S contains a point of the closed subscheme S_0. Hence we can find a specialization $x \rightsquigarrow x_0$ in $|X|$ with $x_0 \in |X_0|$. Choose $u_0 \in U_0$ mapping to x_0. By Decent Spaces, Lemma 0B7W (or by Decent Spaces, Lemma 03K2 which applies directly to étale morphisms) we find a specialization $u \rightsquigarrow u_0$ in U such that u maps to x. We may lift $\overline{x}$ to a geometric point $\overline{u}$ of U lying over u. Then we have $E_{\overline{x}} = (E|_U)_{\overline{u}}$.

Write $U = \mathrm{Spec}(A)$. Then A is a flat, finitely presented R-algebra which is a quotient of a polynomial R-algebra in d-variables. The restriction $E|_U$ corresponds (by Derived Categories of Spaces, Lemmas 08JL, 071Q, and 08HE and Derived Categories of Schemes, Lemma 06Z0 and 08E7) to a pseudo-coherent object K of $D(A)$. Observe that E_0 corresponds to $K \otimes^{\mathbf{L}}_A A/IA$. Let $\mathfrak{q} \subset \mathfrak{q}_0 \subset A$ be the prime ideals corresponding to $u \rightsquigarrow u_0$. Then

$$E_{\overline{x}} = (E|_U)_{\overline{u}} = E_u \otimes^{\mathbf{L}}_{\mathcal{O}_{U,u}} \mathcal{O}_{U,\overline{u}} = K_{\mathfrak{q}} \otimes^{\mathbf{L}}_{A_{\mathfrak{q}}} A^{sh}_{\mathfrak{q}}$$

(some details omitted). Since $A_{\mathfrak{q}} \to A^{sh}_{\mathfrak{q}}$ is flat, the tor amplitude of this as an R-module is the same as the tor amplitude of $K_{\mathfrak{q}}$ as an R-module (More on Algebra, Lemma 0DJF). Also, $K_{\mathfrak{q}}$ is a localization of $K_{\mathfrak{q}_0}$. Hence it suffices to show that $K_{\mathfrak{q}_0}$ has tor amplitude in $[a-d, b]$ as a complex of R-modules.

Let $I \subset \mathfrak{p}_0 \subset R$ be the prime ideal corresponding to $f(x_0)$. Then we have

$$\begin{aligned} K \otimes^{\mathbf{L}}_R \kappa(\mathfrak{p}_0) &= (K \otimes^{\mathbf{L}}_R R/I) \otimes^{\mathbf{L}}_{R/I} \kappa(\mathfrak{p}_0) \\ &= (K \otimes^{\mathbf{L}}_A A/IA) \otimes^{\mathbf{L}}_{R/I} \kappa(\mathfrak{p}_0) \end{aligned}$$

the second equality because $R \to A$ is flat. By our choice of a, b this complex has cohomology only in degrees in the interval $[a, b]$. Thus we may finally apply More on Algebra, Lemma 0DJH to $R \to A$, $\mathfrak{q}_0$, $\mathfrak{p}_0$ and K to conclude. □